Logit Adjustment Meets Transfer Learning: Balanced Skin-Lesion Classification on DermaMNIST

Deep Learning Challenge — SRH University of Applied Sciences

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1. PROBLEM STATEMENT & DATASET

HAM10000 / MEDMNIST V2

Dermatological classification algorithms often suffer from severe class imbalance. In **DermaMNIST** (28×28 RGB images), harmless moles constitute 67% of training data. A baseline model predicting exclusively "nevus" achieves 67% accuracy but detects 0% melanomas.

Core Challenge: 59:1 imbalance ratio between Melanocytic nevi (4,693 images) and Dermatofibroma (80 images). Evaluation mandates **Balanced Accuracy** (mean sensitivity across all 7 classes).

Code	Class Description	Train	Val	Test	Share
akiec	Actinic keratoses	228	33	66	3.3%
bcc	Basal cell carcinoma	359	52	103	5.1%
bkl	Benign keratosis-like	769	110	220	11.0%
df	Dermatofibroma	80	12	23	1.1%
mel	Melanoma (Malignant)	779	111	223	11.1%
nv	Melanocytic nevi (Moles)	4,693	671	1,341	67.0%
vasc	Vascular lesions	99	14	29	1.4%

* Official dataset splits total 10,015 dermatoscopy images (7,007 train / 1,003 val / 2,005 test).

2. CONTROLLED METHODOLOGY

SINGLE-FACTOR EVALUATION

We tested **10 individual improvements** against the starter CNN baseline (16,071 parameters). Every experiment was trained across 3 random seeds (0, 1, 2) using fixed budgets to isolate gains.

$$\text{Training Loss: } L = \text{CrossEntropy}(z + \tau \cdot \log(\pi), y) \;\mid\; \text{Test: raw logits } z$$

- **Logit Adjustment (E2):** Enforces Fisher-consistency for balanced error by adding prior log-frequencies $\tau \log(\pi)$ during training ($\tau = 1.0$).
- **Pretrained Backbone (E7):** ImageNet ResNet-18 (11.2M params) with internal bilinear upsampling to 112×112 pixels.
- **Data Augmentations (E3):** Random horizontal/vertical flips and 90° rotations reflecting unconstrained dermatoscopy capture.
- **Final Stack (S2):** Combines E2 + ResNet-18 + Spatial Augmentations. Final inference averages logits across a 3-seed ensemble.

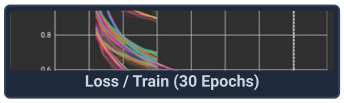
10	3 Seeds	2,000
SINGLE CHANGES	PER EXPERIMENT	BOOTSTRAP CI SAMPLES

3. EXPERIMENTAL RESULTS & DYNAMICS

95% PAIRED BOOTSTRAP CI

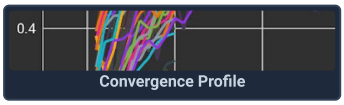
Model / Experiment Variant	Val Bal. Acc.	Δ vs Baseline [95% CI]
Baseline (Starter CNN)	0.461 ± 0.011	Reference Baseline
E1: Class-Weighted CE	0.515 ± 0.016	+0.054 [+0.023, +0.092]
E2: Logit-Adjusted Loss (τ=1.0)	0.532 ± 0.031	+0.071 [+0.029, +0.118]
E7: Pretrained ResNet-18	0.493 ± 0.051	+0.032 [-0.011, +0.075]
Stack S2: S1 + Flips/Rotations	**0.590 ± 0.017**	**+0.129 [+0.078, +0.181]**
Final Stack S2 (3-Seed Ensemble)	0.609	Ensemble Boost

Training Dynamics & Convergence Logs (TensorBoard):









Test Evaluation: Final S2 Ensemble achieved **0.632 Balanced Acc.** vs 0.448 Baseline (**Δ +0.184**, 95% CI [+0.136, +0.232]).

4. PER-CLASS SHIFT & DEPLOYMENT

TORCHSCRIPT & GRADIO

Per-Class Sensitivity Shift (Test Split):

Class	Baseline	Final Model	Sensitivity Gain
df (Dermatofibroma)	0.09	0.57	+0.48 Gain
mel (Melanoma)	0.34	0.64	+0.30 Gain
vasc (Vascular)	0.45	0.93	+0.48 Gain
akiec (Actinic Ker.)	0.33	0.52	+0.19 Gain
nv (Harmless Moles)	0.91	0.73	-0.18 Trade-off

Deployment Pipeline: Exported as a single `model.pt` file using TorchScript serialization. Inputs (28×28×3) are automatically normalized and upsampled on-the-fly. Deployed with a interactive **Gradio Web UI** for top-3 class confidence visualization.

Conclusion: Logit adjustment combined with transfer learning effectively solves severe class imbalance, substantially improving life-saving melanoma detection sensitivity (+0.30).